

# ABDUL BASSET CHOUMAN

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■ VANCOUVER, BC, V5R 6H4, CANADA

## PROFILE

Senior Backend Engineer specializing in Java-based distributed systems and microservices. Extensive experience designing and building scalable, high-throughput backend services for data-intensive applications. Strong expertise in API design, event-driven architecture, and cloud-native systems on AWS. Proven ability to optimize performance, ensure data integrity, and design systems for efficient data retrieval and querying at scale. Skilled in Java frameworks such as Quarkus, Spring Boot and Vert.x, with a focus on building robust, maintainable code. Experienced in implementing secure authentication and authorization using Keycloak, and integrating with various identity providers. Adept at collaborating in agile teams to deliver high-quality software solutions that meet business needs.

## CORE TECHNICAL SKILLS

**Backend Languages:** Java, JavaScript (Node.js), TypeScript

**Frameworks:** Quarkus, Spring Boot, Vert.x, Express.JS

**Architecture & APIs:** Microservices, REST APIs, Event-Driven Architecture, OpenAPI/Swagger, High-Throughput Systems, Data-Intensive Applications

**Databases & Caching:** PostgreSQL, MongoDB, Cassandra, Redis, MySQL, MariaDB

**Messaging And Streaming:** Kafka

**Search & Indexing:** Query Optimization, Data Retrieval Patterns

**Cloud & DevOps:** AWS, Kubernetes, Docker, Jenkins, CI/CD

**Security & Identity:** Keycloak, OAuth2, RBAC

**Other:** Linux, Shell Scripts

## PROFESSIONAL EXPERIENCE

### Senior Software Engineer | Willow Laboratories | Vancouver | Mar 2025 - PRESENT

- Designed and implemented high-throughput backend microservices using Java (Quarkus) to process real-time healthcare data streams.
- Built secure and scalable identity and access management flows using Keycloak, supporting multi-system integrations.
- Designed and documented RESTful APIs with a focus on scalability, maintainability, and compliance with healthcare data standards.
- Optimized database performance and reduced service latency through query tuning, indexing strategies, and efficient data access patterns.
- Improved system observability by integrating monitoring and logging solutions for better reliability and debugging.
- Collaborated within an Agile Scrum team, contributing to system design discussions, code reviews, and backend architecture decisions.

### Senior Engineer, Cloud & Web Systems Software | Masimo | Vancouver | Sep 2022 - Mar 2025

- Architected distributed microservices using Spring Boot and Vert.x for real-time systems handling large volumes of device data.
- Built backend services for Over-the-Air (OTA) firmware updates for IoT/Bluetooth devices, handling asynchronous communication and event-driven data processing.
- Designed high-performance APIs and optimized backend data flows for real-time clinical monitoring systems, improving throughput and query efficiency.
- Implemented CI/CD pipelines using Jenkins and Docker, enabling automated builds, testing, and deployments to AWS.
- Developed internal tools and frameworks (Java, Groovy) to automate testing and infrastructure replication.
- Contributed to system architecture design with a focus on scalability, reliability, and data integrity in regulated environments.
- Documented system architecture and technical designs to align cross-team collaboration and ensure maintainability.

## **Software Engineer | Vistory | Paris | Sep 2017 - Jul 2022**

- Designed and implemented backend services using Node.js and Express.js for a 3D printing management platform.
- Modeled and optimized relational databases (MariaDB) and caching layers (Redis), improving application performance and scalability.
- Built RESTful APIs supporting high-volume user interactions, optimizing data access and retrieval across large datasets.
- Developed and deployed microservices using Docker and Jenkins-based CI/CD pipelines.
- Implemented blockchain-based features (ERC721 smart contracts) for secure digital asset ownership.

## **SYSTEM DESIGN & BACKEND EXPERTISE**

- Distributed systems design
- Event-driven architecture (Kafka)
- API design, versioning, and documentation
- Database optimization and indexing strategies
- High-availability and fault-tolerant system design

## **EDUCATION**

Bachelor's Degree in Computer Science | American University of Science and Technology | Beirut | Feb 2010 - Feb 2017

## **CERTIFICATIONS & COURSES**

Linux Professional Institute (LPIC-1) | AUST | Beirut | Feb 2016 - Jul 2016

Cisco Certified Network Associate (CCNA) | AUST | Beirut | Feb 2015 - Jul 2015

## **PERSONAL PROJECTS**

### **GrapeCastle:**

- Designed a microservices-based video streaming platform using event-driven architecture with Kafka.
- Implemented centralized authentication and authorization using Keycloak across distributed services.
- Built real-time communication systems using messaging services and WebSocket-based interactions.
- Designed scalable backend services with PostgreSQL and service-to-service communication.

### **Perfect Resume:**

- Developed a RESTful API using Node.js to generate structured, design-friendly resumes.

### **Solid Deployer:**

- Built a Node.js-based engine to compile and deploy Solidity smart contracts on Ethereum.

## **LANGUAGES**

English

## **REFERENCES**

Available upon request